Part I: Cloud Computing Fundamentals

Cloud computing is the on-demand delivery of IT resources over the Internet with pay-as-you-go pricing. Core service models include Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS). Virtualization enables multiple virtual machines to run on a physical server.

LECTURE NOTES - SCANNED COPY

Handwritten Diagram: Virtual Machine vs Container Architecture

Hypervisor Type 1 runs directly on hardware.

Containers share the host kernel while providing process isolation.

Part II: Distributed Systems & Consensus

Consensus algorithms allow a collection of machines to work as a coherent group that can survive failures. The Raft consensus algorithm is designed to be easy to understand and decomposes consensus into leader election, log replication, and safety.

Lecture Slide Note: CAP Theorem Tradeoffs

Consistency, Availability, Partition Tolerance.

In the presence of a network partition, a distributed system must choose between Consistency and Availability.